Interfaces

Tags	Interface collection and storage, Cannot subtract those objects, collect_if.py, do_validation, derive_if_rates, raw_time, post_collect_pdu_packing_enabled, IPAddressError: Address passed was not valid, derive_if_rates, raw_time
Team Owner	The Collectables
Status	ACTIVE
Last Audit	Mar 8, 2021

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Introduction

Diagnosis and Mitigation

Disabling a long list of interfaces by device group

If you have a numeric list of device group ids to work with you could try this:

```
# you can check the list of interfaces with this
select di.did, dg2d.dgid
from master_dev.device_interfaces di
join master_dev.device_groups_to_devices as dg2d on (dg2d.did=di.did)
where dg2d.dgid in (1,2,3);

# and disable the list with this
update master_dev.device_interfaces di
join master_dev.device_groups_to_devices as dg2d on (dg2d.did=di.did)
set state=0
where dg2d.dgid in (1,2,3);
```

and if you have a list of device group names you could do this:

```
# you can check the list of interfaces with this
select di.did, dg2d.dgid
from master_dev.device_interfaces di
join master_dev.device_groups_to_devices as dg2d on (dg2d.did=di.did)
join master_dev.device_groups as dg on (dg.dgid=dg2d.dgid)
where dg.name in ("a","b","c");

# and disable the list with this
update master_dev.device_interfaces di
join master_dev.device_groups_to_devices as dg2d on (dg2d.did=di.did)
join master_dev.device_groups as dg on (dg.dgid=dg2d.dgid)
set state=0
where dg.name in ("a","b","c");
```

```
```

Re-enabling would be a matter of changing set state=0 to set state=1.

Theory

Orphaned Interfaces

Post Mortem Articles

2024-07-31 EM-66209: aws-tcs-ddp-sl1e-de-02-prod - nleidm125 medium frequency is 156095 behind (threshold: 120000)

ts_guide_to_jira_link

Link: EM-66209: aws-tcs-ddp-sl1e-de-02-prod - nleidm125 medium frequency is 156095 behind (threshold: 120000) CLOSED

From the description:

```
Observed Behavior: Increasing MF behind, large master_dev.device_interfaces where did=0

Expected Behavior: No Increasing MF behind & large master_dev.device_interfaces where did=0

Summary of troubleshooting steps taken so far and the current status:

SaaS stack has increasing MF behind with busy RDS, the below query was observed to be taking a while.

SELECT COUNT( DISTINCT main.if_id) FROM master_dev.device_interfaces main
LEFT JOIN (SELECT di.if_id, di.did, IF(di.roa_id=0,ld.roa_id,di.roa_id) as roa_id
FROM master_dev.device_interfaces di
INNER JOIN master_dev.legend_device ld ON (di.did=ld.id)
) iforg  ON (main.if_id=iforg.if_id)
WHERE (main.roa_id IN (17) OR main.did IN (SELECT id FROM master_dev.legend_device ld WHERE ld.id = main.did AND ld.roa_id IN (17))) LIMIT 1

Can Support or SRE be approved to clean up the table?

DELETE from `master_dev`.`device_interfaces` WHERE did = 0 LIMIT 200000

MySQL [(none)]> select count(*) from master_dev.device_interfaces where did=0;
+----------+
| count(*) |
+----------+
|   201583 |
+----------+
1 row in set (0.074 sec)

MySQL [(none)]> select count(*) from master_dev.device_interfaces;
+----------+
| count(*) |
+----------+
|   714666 |
```

```
+----------+
1 row in set (0.191 sec)

Slack channel https://sciencelogic.slack.com/archives/C8B2JLWMR/p1722339503764299
```

From this comment: EM-66209: aws-tcs-ddp-sl1e-de-02-prod - nleidm125 medium frequency is 156095 behind (threshold: 120000) CLOSED

```
The Collectables:

I am assigning this to your group to answer the questions around interfaces with did=0. Specifically: is it OK to delete them?  Please tag me in your reply, as we will probably want to reassign this case
to another team (probably Proc and Roll) depending on what we find out about the DDGU process and whether its the main problem here.

From the slack chat:
Something I remembered seeing a while back was the device_interfaces table having too many rows with did=0, so I did a check:
MySQL [(none)]> select count(*) from master_dev.device_interfaces where did=0;
+----------+
| count(*) |
+----------+
|   201583 |
+----------+
1 row in set (0.074 sec)

MySQL [(none)]> select count(*) from master_dev.device_interfaces;
+----------+
| count(*) |
+----------+
|   714666 |
+----------+
1 row in set (0.191 sec)
```

and from this comment: EM-66209: aws-tcs-ddp-sl1e-de-02-prod - nleidm125 medium frequency is 156095 behind (threshold: 120000) CLOSED

```
Collectables:

One of the queries causing problems here looks like it is a query the php API backend uses to retrieve information about interfaces.

This is one of the queries that is wasting a lot of time:
SELECT COUNT( DISTINCT main.if_id) FROM master_dev.device_interfaces main
 LEFT JOIN (SELECT di.if_id, di.did, IF(di.roa_id=0,ld.roa_id,di.roa_id) as roa_id
        FROM master_dev.device_interfaces di
        INNER JOIN master_dev.legend_device ld ON (di.did=ld.id)
        ) iforg  ON (main.if_id=iforg.if_id)
WHERE (main.roa_id IN (17) OR main.did IN (SELECT id FROM master_dev.legend_device ld WHERE ld.id = main.did AND ld.roa_id IN (17))) LIMIT 1

I believe this is the source code it is originating from: https://github.sciencelogic.com/SL1/em7_v8/blob/e318dc85b82595368532819d38036490100a3c17/G3code/gui/is/classes/features/ftr_interface.inc
Its from this file G3code/gui/is/classes/features/ftr_interface.inc in the main repo:
...
```

```
class ftr_interface extends base_ftr_entity {
...
  protected function _api_idx_args($fields) {
    $args = parent::{__FUNCTION__}($fields);
    $args['table_spec'] += [
      'ld'=>['master_dev.legend_device', 'main.did=ld.id'],
      'iforg'=>[
        '(SELECT di.if_id, di.did, IF(di.roa_id=0,ld.roa_id,di.roa_id) as roa_id
        FROM master_dev.device_interfaces di
        INNER JOIN master_dev.legend_device ld ON (di.did=ld.id)
        )',
        'main.if_id=iforg.if_id'
      ],
      'org'=>['master_biz.organizations', 'iforg.roa_id=org.roa_id'],

      'tag_map' => ["(SELECT it_map.if_id, tags.tag_id, tags.tag_name  FROM
      master_dev.device_interface_tags_map as it_map JOIN
      master_dev.device_interface_tags as tags ON it_map.tag_id = tags.tag_id
      )", 'main.if_id=tag_map.if_id']
    ];

    $args['pub_to_tablefield']['tag/id'] = 'tag_map.tag_id';
    $args['pub_to_tablefield']['tag/name'] = 'tag_map.tag_name';
    $args['pub_to_tablefield']['device/name'] = 'ld.device';
    $args['pub_to_tablefield']['organization'] = 'iforg.roa_id';
    $args['pub_to_tablefield']['organization_device'] = 'ld.roa_id'; // expose the device's org as a filter field
    $args['pub_to_tablefield']['organization_interface'] = 'main.roa_id'; // expose the interface's org as a filter field
    $args['pub_to_tablefield']['organization/company'] = 'org.company';
    // without one field from the iforg table in the query, the framework will skip this filter.
    // this overrides that behavior, causing a join regardless if a field from the table is referenced.
    $args['base_filters'] += [
      'org_stuff'=>[ 'rawsql', ['main', 'iforg'], NULL, NULL],
    ];

    return $args;
  }

I believe this code is used by the API to retrieve information about interfaces.
If  a dashboard were using this API endpoint to find a list of interfaces, that could account for why this query is being produced.
```

2024-07-11 EM-65798: KPN Stack 33_ EM7 critical event: Unhandled exception during collection, process Data Collection

ts_guide_to_jira_link

Link: 💔EM-65798: KPN Stack 33_ EM7 critical event: Unhandled exception during collection, process Data Collection DONE

From the description:

```
On stack 33 the following critical exception has been active for 4 days now. Could you please investigate and resolve this.
```

```
EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth: Traceback (most recent call last):
File "/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async
File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get raise self._value ChildProcessException: Traceback (most recent call last):
File "/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py", line 43, in new_f File "collect_if.py", line 730, in collect
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if_utils.py", line 1107, in details_postprocessing
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if_utils.py", line 370, in do_validation
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if_utils.py", line 1445, in derive_if_rates KeyError: 'raw_time'

Description
Reason for Escalation:
Engineering assistance needed to diagnose issue in order to file defect
Engineering assistance needed to diagnose issue in order to find workaround
Which troubleshooting doc(s) were reviewed: NA
Customer Timezone: Anytime
Versioning & Environment: SL1 Build: 12.1.1
Architecture: SL1 Extended Architecture
Remote Access: SaaS-AWS
Salesforce Case Number: 00447219
Observed Behavior:Unhandled exception during collection, process Data Collection: Interface Bandwidth
Expected Behavior:Error should not be appeared.
Summary of troubleshooting steps taken so far and the current status:
Customers upgrade their stack from 11.3.1 to 12.1.1 and still seeing below UHE -
1)Customer reported that he got below error
EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth: Traceback (most recent call last):
File "/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async
File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get raise self._value ChildProcessException: Traceback (most recent call last):
File "/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py", line 43, in new_f File "collect_if.py", line 730, in collect
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if_utils.py", line 1107, in details_postprocessing
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if_utils.py", line 370, in do_validation
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if_utils.py", line 1445, in derive_if_rates KeyError: 'raw_time'
2)We have verified the process entry in System > Settings > Processes, and both the batch factor and time factor, as well as the frequency, appeared to be correct. We enabled debugging on the process and collected logs.
3) logs(attached) reporting UHE as unsupported operand type(s) for +: 'NoneType' and 'NoneType'
For above error we have checked, high_precision_if_collect value is set to 1 in master.system_custom_config, but customer was admant to change this setting to zero .
4) Upon further investigation into the storage of base metrics on a sample interface_id 43599 on device id 3356, it appears that the interface bandwidth process is not creating entries in the data_if_ext tables.
This issue could be related to a storage problem.
5) Subsequently, we conducted an SNMP walk on the same sample device and discovered that the interface counter is returning as 32-bit, whereas on SL1, the interface counter is 64-bit.
```

```
As a suggestion, we proposed disabling
Interface Speed/Counter Type Update and then retrying. However, the customer remained insistent on finding a root fix for this issue..
Re-opening of this Jira case (https://sciencelogic.atlassian.net/browse/EM-64212#icft=EM-64212)
```

The previous case, being reopened here, has a separate section below.

I am guessing that the UHE is being raised from the second to last line of code below:

```
def derive_if_rates(
    dbc, logger, current_raw, last_irregular, last_regular, use_raw_times=True  # interface object
):
    """
    returns [computed rates], irregular_to_stash, regular_to_stash

    It's IF collect's job to stash the values and supply them on the next
    iteration

    Rate data points are in the form of:
    (
        timestamp,
        {
            "ifInOctets": rate,
            "ifOutOctets": rate,
            "ifInErrors": rate,
            "ifOutErrors": rate,
            "ifInDiscards": rate,
            "ifOutDiscards": rate
        }
    )
    """
    time_key = "raw_time" if use_raw_times else "poll_time"
    if_id = current_raw.if_id

    new_data = {f: current_raw.properties[f] for f in ALL_COUNTERS}
    new_ts = current_raw.properties[time_key]
    new_datapoint = (new_ts, new_data)
```

2024-07-10 EM-65760: Critical System log messages about Interface Bandwidth collection and Address passed not valid

Link: EM-65760: Critical System log messages about Interface Bandwidth collection and Address passed not valid CLOSED

From the description:

```
Observed Behavior:UHE on interface bandwidth process "Address Passed not valid"

Expected Behavior: Should not be UHE error if the IP address looks fine

Summary of troubleshooting steps taken so far and the current status:
1)Enabled debug for 'Data Collection: Interface Bandwidth Process' and encountered this error:
```

```
2024-07-10T00:15:04.425426+00:00 localhost silo.if_collect.12[2105]: CRITICAL shared._run_collect_async 230 Unhandled exception during collection, process Data Collection: Interface Bandwidth:#012Traceback (most recent call last):#012  File "/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async#012  File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get#012    raise self._value#012ChildProcessException: Traceback (most recent call last):#012  File "/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py", line 43, in new_f#012  File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 732, in collect#012 File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 708, in collect_interfaces#012  File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 644, in collect_interfaces_legacy#012  File "/tmp/pip-install-PbZ5ZM/sl-network/sl_network/network.py", line 120, in _init_#012IPAddressError: Address passed was not valid
2)Ran the query below to check for issues with device_ip_addr, but did not find any invalid IPs. The subnet mask is 64 for IPv6, which I believe is correct.
SELECT * FROM master_dev.device_ip_addr WHERE ip=' %' OR ip LIKE '% ' or netmask LIKE ' %' OR netmask LIKE '% ' OR (NOT IS_IPV4(ip)) AND (NOT IS_IPV6(ip));
3)I checked for bad IPs in master_dev.legend_device and master_dev.device_ports but did not find any. Since I have no further leads, I need your expertise to identify the root cause and resolve the issue.
4)attaching all the logs and screenshots
```

The quoted UHE, edited to be human readable, is:

```
2024-07-10T00:15:04.425426+00:00 localhost silo.if_collect.12[2105]: CRITICAL shared._run_collect_async 230 Unhandled exception during collection, process Data Collection: Interface Bandwidth:#012Traceback (most recent call last):#012
 File "/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async#012
 File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get#012    raise self._value#012ChildProcessException: Traceback (most recent call last):#012
 File "/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py", line 43, in new_f#012
 File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 732, in collect#012
 File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 708, in collect_interfaces#012
 File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 644, in collect_interfaces_legacy#012
 File "/tmp/pip-install-PbZ5ZM/sl-network/sl_network/network.py", line 120, in _init_#012IPAddressError: Address passed was not valid

```

2024-07-03 EM-65671: post_collect_pdu_lacking SIGTERMS on collector

ts_guide_to_jira_link

Link: EM-65671: post_collect_pdu_lacking SIGTERMS on collector CLOSED

From the description:

```
Observed Behavior: PDU_Packing sigterms for interface collection, even though PDU Packing is disabled.
Expected Behavior: No sigterms should be present.
Summary of troubleshooting steps taken so far and the current status:

Customer is having all interface collections affected from the following sigterm:

```

```
EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth: Traceback (most recent call last):
 File "/tmp/pip-install-Mq7wCs/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async
 File "/usr/lib64/python2.7/multiprocessing/pool.py", line 572, in get raise self._value ChildProcessException: Traceback (most recent call last):
 File "/tmp/pip-install-Mq7wCs/sl-process/sl_process/exceptions.py", line 43, in new_f
 File "/tmp/pip-install-Mq7wCs/sl-interface/sl_interface/collect_if.py", line 1021, in collect_service
 File "/tmp/pip-install-Mq7wCs/sl-interface/sl_interface/collect_if.py", line 979, in collect_interfaces
 File "/tmp/pip-install-Mq7wCs/sl-interface/sl_interface/collect_if.py", line 515, in post_collect_pdu_packing_enabled KeyError: '.1.3.6.1.2.1.31.1.1.1.2.41'
Interface version for collector/DB:
interface_version.png
master.definitions_if_metric_oids
snmpwalk of device customer believes is causing the issue. They have changed the polling to 120 minutes, sigterms have reduced

Device and interfaces

interfaces.png
device.png

Thank you.
```

2024-04-17 EM-64487: Critical System Event

ts_guide_to_jira_link

Link: 💔EM-64487: Critical System Event DONE

From the description:

```
Customer getting UHE for Data Collection: Interface Bandwidth on multiple collectors [ MID: 11, 29, 68, 70]
Error:
EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth: Traceback (most recent call last):
File "/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async
File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get raise self._value ChildProcessException: Traceback (most recent call last):
File "/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py", line 43, in new_f
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 1021, in collect_service
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 979, in collect_interfaces
File "/tmp/pip-install-PbZ5ZM/sl-interface/sl_interface/collect_if.py", line 515, in post_collect_pdu_packing_enabled KeyError: '.1.3.6.1.2.1.31.1.1.1.2.2'
We are seeing KeyError for multiple oids.
We have checked and it does not seem the pud is enabled for the device classes we checked.
1.3.6.1.2.1.31.1.1.1.8  - IF-MIB
Object Type: Counter64
Object Name: ifHCInMulticastPkts
.1.3.6.1.2.1.31.1.1.1.2 -   IF-MIB
Object Type: Counter32
```

```
Object Name: ifInMulticastPkts
Attached the if_collect logs from the collectors
```

From this comment 💔 EM-64487: Critical System Event DONE

```
Hello @Ketan Gharate after notice that 12.1.1 have different sl_interfaces versions we updated the collect_if.py dropfile for the different versions. in attached zip you will find a collect_if.2_new.pyc and collect_if.3.pyc, follow next steps to setup:

For python2 (collect_if.2_new.pyc)
    cp /tmp/collect_if.2_new.pyc /opt/em7/lib/python/sl_interface/collect_if.pyc
    chgrp s-em7-srv /opt/em7/lib/python/sl_interface/collect_if.pyc
    chmod 640 /opt/em7/lib/python/sl_interface/collect_if.pyc
    mv /opt/em7/lib/python/sl_interface/collect_if.pyc /opt/em7/lib/python/sl_interface/collect_if.pyc

For python3 (collect_if.3.pyc)
    cp /tmp/collect_if.3.py /opt/em7/lib/python3.6/sl_interface/collect_if.py
    chmod 744 /opt/em7/lib/python3.6/sl_interface/collect_if.py
    mv /opt/em7/lib/python3.6/sl_interface/collect_if.pyc /opt/em7/lib/python3.6/sl_interface/collect_if.pyc
    cp /tmp/collect_if.3.py /opt/em7/lib/python3.9/sl_interface/collect_if.py
    chmod 744 /opt/em7/lib/python3.9/sl_interface/collect_if.py
    mv /opt/em7/lib/python3.9/sl_interface/collect_if.pyc /opt/em7/lib/python3.9/sl_interface/collect_if.pyc

collect_if.zip
05 Jun 2024, 04:40 PM
```

It looks like this case was resolved with a drop file.

2024-04-15 EM-64212: STACK 33-EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth

ts_guide_to_jira_link

Link: EM-64212: STACK 33-EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth CLOSED

From the description:

```
Observed Behavior:Unhandled exception during collection, process Data Collection: Interface Bandwidth

Expected Behavior:Error should not be appeared.

Summary of troubleshooting steps taken so far and the current status:

1)Customer reported that he got below error
EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth: Traceback (most recent call last):
File "/opt/em7/backend/silo_collect/shared.py", line 167, in _run_collect_async
File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get raise self._value ChildProcessException: Traceback (most recent call last):
File "/opt/em7/backend/silo_common/misc.py", line 568, in new_f
File "/opt/em7/backend/silo_collect/collect_if.py", line 578, in collect
```

```
File "/opt/em7/backend/silo_collect/collect_if_utils.py", line 836, in details_postprocessing
File "/opt/em7/backend/silo_collect/collect_if_utils.py", line 303, in do_validation
File "/opt/em7/backend/silo_collect/collect_if_utils.py", line 1142, in derive_if_rates
KeyError: 'raw_time'

2)We have verified the process entry in System > Settings > Processes, and both the batch factor and time factor, as well as the frequency, appeared to be correct. We enabled debugging on the process and collected logs.
3) logs(attached) reporting UHE as unsupported operand type(s) for +: 'NoneType' and 'NoneType'
For above error we have checked, high_precision_if_collect value is set to 1 in master.system_custom_config, but customer was admant to change this setting to zero .
4) Upon further investigation into the storage of base metrics on a sample interface_id 43599 on device id 3356, it appears that the interface bandwidth process is not creating entries in the data_if_ext tables. This issue could be related to a storage problem.
5) Subsequently, we conducted an SNMP walk on the same sample device and discovered that the interface counter is returning as 32-bit, whereas on SL1, the interface counter is 64-bit. As a suggestion, we proposed disabling Interface Speed/Counter Type Update and then retrying. However, the customer remained insistent on finding a root fix for this issue..
6)One observation the customer shared is that they began receiving the UHE report after applying the drop file from https://sciencelogic.atlassian.net/browse/EM-61728#icft=EM-61728.
7)attaching all the screenshot and logs
```

2024-04-09 EM-64212: STACK 33-EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth

ts_guide_to_jira_link

Link: EM-64212: STACK 33-EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth CLOSED

From the description:

```
Observed Behavior:Unhandled exception during collection, process Data Collection: Interface Bandwidth
Expected Behavior:Error should not be appeared.
Summary of troubleshooting steps taken so far and the current status:
1)Customer reported that he got below error
EM7 critical event: Unhandled exception during collection, process Data Collection: Interface Bandwidth: Traceback (most recent call last):
File "/opt/em7/backend/silo_collect/shared.py", line 167, in _run_collect_async
File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get raise self._value
ChildProcessException: Traceback (most recent call last):
File "/opt/em7/backend/silo_common/misc.py", line 568, in new_f File "/opt/em7/backend/silo_collect/collect_if.py", line 578, in collect
File "/opt/em7/backend/silo_collect/collect_if_utils.py", line 836, in details_postprocessing
File "/opt/em7/backend/silo_collect/collect_if_utils.py", line 303, in do_validation
File "/opt/em7/backend/silo_collect/collect_if_utils.py", line 1142, in derive_if_rates
KeyError: 'raw_time'
```

2024-03-26 EM-63991: Cisco devices Interface data collection issue

ts_guide_to_jira_link

Link: EM-63991: Cisco devices Interface data collection issue CLOSED

From the description:

```
Observed Behavior: Interfaces on Cisco devices for one organization do not seem to be collecting data.
Expected Behavior: All interfaces should correctly collect data.
```

2024-02-06 EM-63211: Interface Events With No Description

ts_guide_to_jira_link

Link: EM-63211: Interface Events With No Description CLOSED

From the description

```
Observed Behavior: The event interface name are not updating properly and resulting to have variables instead of values.

Expected Behavior: The event interface name has to be substituted with variables configured in behavior setting and show actual values.

Summary of troubleshooting steps taken so far and the current status:

Cross verified the settings in behavior page.
Cross verified the interface configurations.
Did a snmp walk for the devices that had issues and the results were verified.
collected the device information that had these type of issues.
```

2023-10-26 EM-61788 Upgrading SL1 caused Event Flood with "Cannot subtract those objects - 'other' is not an object of type 'interface'"

ts_guide_to_jira_link

Link: EM-61788: Upgrading SL1 caused Event Flood with "Cannot subtract those objects - 'other' is not an object of type 'interface'" CLOSED

This was fixed in this bug story: EM-56169: OL7 12.1.0 patch. Unhandled exception during collection, process Data Collection: Interface Bandwidth. ('bad object', "Cannot subtract those objects - 'other' is not an object of type 'interface'") CLOSED

but occurred later.

The UHE quoted in the bug story is:

```
File "/opt/em7/lib/python/sl_collect/shared.py", line 174, in _run_collect_async self.devices[device] = result.get()
File "/usr/lib64/python2.7/multiprocessing/pool.py", line 554, in get raise self._value ChildProcessException: Traceback (most recent call last):
File "/opt/em7/lib/python/sl_process/exceptions.py", line 43, in new_f res = func(*args, **kwargs)
File "/opt/em7/lib/python/sl_interface/collect_if.py", line 734, in collect details, mid, dbc_local, logger, start_time, start_clock, sm, debug, errors
```

```
File "/opt/em7/lib/python/sl_interface/collect_if_utils.py", line 1099, in details_postprocessing do_validation(details["result_if_list"], device_obj, dbc_local, details, logger)
File "/opt/em7/lib/python/sl_interface/collect_if_utils.py", line 437, in do_validation delta = interface_for_delta - last_interface
File "/opt/em7/lib/python/sl_interface/interface.py", line 265, in __sub__ "Cannot subtract those objects - 'other' is not an object of type 'interface'", TypeError: ('bad object', "Cannot subtract those objects - 'other' is not an object of type 'interface'")

```

Silo Log Messages

Introduction

Analysis Script

Log Pattern Analysis

main process

stanza:a complete run

```
"2020-06-01 12:31:06,148","DEBUG","449","12","proc_pda","256","Got proc virtual memory limit from silo.conf:None"
"2020-06-01 12:31:06,148","DEBUG","449","12","proc_pda","268","No or Incorrect virtual memory limit from silo.conf:proc_id=12 : None"
"2020-06-01 12:31:06,186","DEBUG","449","12","proc_pda","190","Logging started for Data Collection: Interface Bandwidth"
"2020-06-01 12:31:06,186","DEBUG","449","12","proc_pda","215","importing pda tracker"
"2020-06-01 12:31:06,189","DEBUG","449","12","proc_pda","500","Started tracking process: 75 12 c3890be6a40311ea9a0f00221967da01 "
"2020-06-01 12:31:06,189","DEBUG","449","12","shared","184","Official collection time: 2020-06-01 12:31:00 timestamp:1591014660"
"2020-06-01 12:31:06,190","DEBUG","449","12","proc_pda","374","Trying to get system settings"
"2020-06-01 12:31:06,193","DEBUG","449","12","shared","184","Official collection time: 2020-06-01 12:31:00 timestamp:1591014660"
"2020-06-01 12:31:06,195","DEBUG","449","12","collect_if","1677","collect_if.compile_packet_metrics(): l_mdefs= [Result(metric_id=1, oid_name='ifInUcastPkts', formula='ifInUcastPkts', alert_type=1), Result(metric_id=2, oid_name='ifInMulticastPkts', formula='ifInMulticastPkts', alert_type=1), Result(metric_id=3, oid_name='ifInBroadcastPkts', formula='ifInBroadcastPkts', alert_type=1), Result(metric_id=4, oid_name='ifOutUcastPkts', formula='ifOutUcastPkts', alert_type=1), Result(metric_id=5, oid_name='ifOutMulticastPkts', formula='ifOutMulticastPkts', alert_type=1), Result(metric_id=6, oid_name='ifOutBroadcastPkts', formula='ifOutBroadcastPkts', alert_type=1), Result(metric_id=7, oid_name='ifInUcastPct', formula='(100. * ifInUcastPkts / ifInTotalPkts) if ifInUcastPkts else 0.', alert_type=1), Result(metric_id=8, oid_name='ifInMulticastPct', formula='(100. * ifInMulticastPkts / ifInTotalPkts) if ifInMulticastPkts else 0.', alert_type=1), Result(metric_id=9, oid_name='ifInBroadcastPct', formula='(100. * ifInBroadcastPkts / ifInTotalPkts) if ifInBroadcastPkts else 0.', alert_type=1), Result(metric_id=10, oid_name='ifOutUcastPct', formula='(100. * ifOutUcastPkts / ifOutTotalPkts) if ifOutUcastPkts else 0.', alert_type=1), Result(metric_id=11, oid_name='ifOutMulticastPct', formula='(100. * ifOutMulticastPkts / ifOutTotalPkts) if ifOutMulticastPkts else 0.', alert_type=1), Result(metric_id=12, oid_name='ifOutBroadcastPct', formula='(100. * ifOutBroadcastPkts / ifOutTotalPkts) if
```

```
ifOutBroadcastPkts else 0.', alert_type=1), Result(metric_id=13, oid_name='ifInTotalPkts',
formula='ifInUcastPkts + ifInMulticastPkts + ifInBroadcastPkts + ifInDiscards + ifInErrors',
alert_type=1), Result(metric_id=14, oid_name='ifOutTotalPkts', formula='ifOutUcastPkts +
ifOutMulticastPkts + ifOutBroadcastPkts + ifOutDiscards + ifOutErrors', alert_type=1),
Result(metric_id=15, oid_name='ifInDiscardsPct', formula='(100. * ifInDiscards /
ifInTotalPkts) if ifInDiscards else 0.', alert_type=3), Result(metric_id=16,
oid_name='ifInErrorsPct', formula='(100. * ifInErrors / ifInTotalPkts) if ifInErrors else
0.', alert_type=2), Result(metric_id=17, oid_name='ifOutDiscardsPct', formula='(100. *
ifOutDiscards / ifOutTotalPkts) if ifOutDiscards else 0.', alert_type=3),
Result(metric_id=18, oid_name='ifOutErrorsPct', formula='(100. * ifOutErrors /
ifOutTotalPkts) if ifOutErrors else 0.', alert_type=2), Result(metric_id=19,
oid_name='ifInErrors', formula='ifInErrors', alert_type=2), Result(metric_id=20,
oid_name='ifOutErrors', formula='ifOutErrors', alert_type=2), Result(metric_id=21,
oid_name='ifInDiscards', formula='ifInDiscards', alert_type=3), Result(metric_id=22,
oid_name='ifOutDiscards', formula='ifOutDiscards', alert_type=3), Result(metric_id=23,
oid_name='ifInTotalPct', formula='100. * ifInTotalRate/interfaceSpeed', alert_type=4),
Result(metric_id=24, oid_name='ifOutTotalPct', formula='100. *
ifOutTotalRate/interfaceSpeed', alert_type=4), Result(metric_id=25,
oid_name='ifInUcastRate', formula='ifInUcastPkts/poll_time', alert_type=1),
Result(metric_id=26, oid_name='ifInMulticastRate', formula='ifInMulticastPkts/poll_time',
alert_type=1), Result(metric_id=27, oid_name='ifInBroadcastRate',
formula='ifInBroadcastPkts/poll_time', alert_type=1), Result(metric_id=28,
oid_name='ifOutUcastRate', formula='ifOutUcastPkts/poll_time', alert_type=1),
Result(metric_id=29, oid_name='ifOutMulticastRate', formula='ifOutMulticastPkts/poll_time',
alert_type=1), Result(metric_id=30, oid_name='ifOutBroadcastRate',
formula='ifOutBroadcastPkts/poll_time', alert_type=1), Result(metric_id=31,
oid_name='ifInTotalRate', formula='ifInOctets/(125000*poll_time)', alert_type=4),
Result(metric_id=32, oid_name='ifOutTotalRate', formula='ifOutOctets/(125000*poll_time)',
alert_type=4), Result(metric_id=33, oid_name='ifInOctets', formula='ifInOctets',
alert_type=4), Result(metric_id=34, oid_name='ifOutOctets', formula='ifOutOctets',
alert_type=4)]"
"2020-06-01
12:31:06,197","DEBUG","449","12","collect_if","1685","collect_if.compile_packet_metrics():
l_odefs= [OID(oid_name='ifInBroadcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.3'),
OID(oid_name='ifInDiscards', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.13'),
OID(oid_name='ifInErrors', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.14'),
OID(oid_name='ifInMulticastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.2'),
OID(oid_name='ifInOctets', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.10'),
OID(oid_name='ifInUcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.11'),
OID(oid_name='ifOutBroadcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.5'),
OID(oid_name='ifOutDiscards', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.19'),
OID(oid_name='ifOutErrors', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.20'),
OID(oid_name='ifOutMulticastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.4'),
OID(oid_name='ifOutOctets', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.16'),
OID(oid_name='ifOutUcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.17'),
OID(oid_name='ifInBroadcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.9'),
OID(oid_name='ifInMulticastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.8'),
OID(oid_name='ifInOctets', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.6'),
OID(oid_name='ifInUcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.7'),
OID(oid_name='ifOutBroadcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.13'),
OID(oid_name='ifOutMulticastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.12'),
OID(oid_name='ifOutOctets', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.10'),
OID(oid_name='ifOutUcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.11')]"
"2020-06-01 12:31:06,217","DEBUG","449","12","database","173","QL: query:      SELECT
dim.metric_id, dim.oid_name AS metric_type,
"2020-06-01 12:31:06,217","DEBUG","449","12","database","174","QL: dbc id:
140321213541136"
```

```
"2020-06-01 12:31:06,218","DEBUG","449","12","database","175","QL: query time: 0.000824928283691"
"2020-06-01 12:31:06,218","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:06,218","DEBUG","449","12","database","177","QL: rowcount:   34"
"2020-06-01 12:31:06,223","DEBUG","449","12","database","173","QL: query:      SELECT pia.alert_id, pia.formula, pit.thresh_id,
"2020-06-01 12:31:06,223","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
"2020-06-01 12:31:06,223","DEBUG","449","12","database","175","QL: query time: 0.00320816040039"
"2020-06-01 12:31:06,224","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:06,224","DEBUG","449","12","database","177","QL: rowcount:   36"
"2020-06-01 12:31:06,238","DEBUG","449","12","database","173","QL: query:      SELECT dim.oid_name as metric_type,
"2020-06-01 12:31:06,239","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
"2020-06-01 12:31:06,239","DEBUG","449","12","database","175","QL: query time: 0.00142192840576"
"2020-06-01 12:31:06,239","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:06,240","DEBUG","449","12","database","177","QL: rowcount:   12"
"2020-06-01 12:31:06,838","DEBUG","449","12","database","173","QL: query:      SELECT ito.xid as did, ito.yid as if_id,
"2020-06-01 12:31:06,838","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
"2020-06-01 12:31:06,838","DEBUG","449","12","database","175","QL: query time: 0.596533060074"
"2020-06-01 12:31:06,839","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:06,839","DEBUG","449","12","database","177","QL: rowcount:   7576"
"2020-06-01 12:31:07,061","DEBUG","449","12","database","173","QL: query:      SELECT iao.xid as did, iao.yid as if_id,
"2020-06-01 12:31:07,062","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
"2020-06-01 12:31:07,062","DEBUG","449","12","database","175","QL: query time: 0.138772010803"
"2020-06-01 12:31:07,062","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:07,062","DEBUG","449","12","database","177","QL: rowcount:   7444"
"2020-06-01 12:31:07,134","DEBUG","449","12","database","173","QL: query:      SELECT dev.did,
"2020-06-01 12:31:07,134","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
"2020-06-01 12:31:07,135","DEBUG","449","12","database","175","QL: query time: 0.00914788246155"
"2020-06-01 12:31:07,135","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:07,135","DEBUG","449","12","database","177","QL: rowcount:   25"
"2020-06-01 12:31:07,195","DEBUG","449","12","database","173","QL: query:      SELECT dev.did, dev.ip, dev.hostname, dev.snmp_cred_id,
"2020-06-01 12:31:07,196","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
"2020-06-01 12:31:07,196","DEBUG","449","12","database","175","QL: query time: 0.059417963028"
"2020-06-01 12:31:07,196","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:07,197","DEBUG","449","12","database","177","QL: rowcount:   7"
"2020-06-01 12:31:07,197","DEBUG","449","12","collect_if","1750","Unable to resolve hostname for did:None using recorded ip 10.10.103.22"
"2020-06-01 12:31:07,276","DEBUG","449","12","database","173","QL: query:      SELECT
"2020-06-01 12:31:07,276","DEBUG","449","12","database","174","QL: dbc id: 140321213541136"
```

```
"2020-06-01 12:31:07,276","DEBUG","449","12","database","175","QL: query time: 0.0531249046326"
"2020-06-01 12:31:07,277","DEBUG","449","12","database","176","QL: lastrowid:  0"
"2020-06-01 12:31:07,277","DEBUG","449","12","database","177","QL: rowcount:   0"
"2020-06-01 12:31:07,277","DEBUG","449","12","shared","124","running collect asynchronously"
"2020-06-01 12:31:07,278","DEBUG","449","12","shared","200","collection_process.get_num_spawn: cpu_count = 8"
"2020-06-01 12:31:07,278","DEBUG","449","12","shared","212","collection_process.get_num_spawn: num_devices = 3, min_spawn = 2, max_spawn = 16, throttle = 0"
"2020-06-01 12:31:07,323","DEBUG","449","12","shared","191","collection_process.start_jobservers: num_procs = 2, worker_task_limit = None, number of requests = 3"
"2020-06-01 12:31:07,335","DEBUG","449","12","shared","193","Started 2 worker processes, task limit = None"
"2020-06-01 12:31:07,336","DEBUG","449","12","shared","139","appending task to queue: 7254"
"2020-06-01 12:31:07,336","DEBUG","449","12","shared","139","appending task to queue: 7260"
"2020-06-01 12:31:07,337","DEBUG","449","12","shared","139","appending task to queue: 10178"
"2020-06-01 12:31:08,339","DEBUG","449","12","shared","164","got results for: 7254"
"2020-06-01 12:31:08,340","DEBUG","449","12","shared","164","got results for: 7260"
"2020-06-01 12:31:09,341","DEBUG","449","12","shared","164","got results for: 10178"
"2020-06-01 12:31:10,344","DEBUG","449","12","collect_if","1670","interface collection complete:  processed count: 7, error count 0"
"2020-06-01 12:31:10,346","DEBUG","449","12","proc_pda","528","tracking update: 75 12 c3890be6a40311ea9a0f00221967da01 "
"2020-06-01 12:31:10,349","DEBUG","449","12","proc_pda","545","Stopped tracking process: 75 12 c3890be6a40311ea9a0f00221967da01 "
"2020-06-01 12:31:10,350","DEBUG","449","12","proc_pda","307","Logging stopped for Data Collection: Interface Bandwidth"


```

stanza:logging started

```
2020-06-01 12:31:06,148","DEBUG","449","12","proc_pda","256","Got proc virtual memory limit from silo.conf:None"
"2020-06-01 12:31:06,148","DEBUG","449","12","proc_pda","268","No or Incorrect virtual memory limit from silo.conf:proc_id=12 : None"
"2020-06-01 12:31:06,186","DEBUG","449","12","proc_pda","190","Logging started for Data Collection: Interface Bandwidth"
"2020-06-01 12:31:06,186","DEBUG","449","12","proc_pda","215","importing pda tracker"
"2020-06-01 12:31:06,189","DEBUG","449","12","proc_pda","500","Started tracking process: 75 12 c3890be6a40311ea9a0f00221967da01 "
"2020-06-01 12:31:06,189","DEBUG","449","12","shared","184","Official collection time: 2020-06-01 12:31:00 timestamp:1591014660"
"2020-06-01 12:31:06,190","DEBUG","449","12","proc_pda","374","Trying to get system settings"
"2020-06-01 12:31:06,193","DEBUG","449","12","shared","184","Official collection time: 2020-06-01 12:31:00 timestamp:1591014660"
"2020-06-01 12:31:06,195","DEBUG","449","12","collect_if","1677","collect_if.compile_packet_metrics(): l_mdefs= [Result(metric_id=1, oid_name='ifInUcastPkts', formula='ifInUcastPkts', alert_type=1), Result(metric_id=2, oid_name='ifInMulticastPkts', formula='ifInMulticastPkts', alert_type=1), Result(metric_id=3, oid_name='ifInBroadcastPkts', formula='ifInBroadcastPkts', alert_type=1), Result(metric_id=4, oid_name='ifOutUcastPkts', formula='ifOutUcastPkts', alert_type=1), Result(metric_id=5, oid_name='ifOutMulticastPkts', formula='ifOutMulticastPkts',
```

```
alert_type=1), Result(metric_id=6, oid_name='ifOutBroadcastPkts',
formula='ifOutBroadcastPkts', alert_type=1), Result(metric_id=7, oid_name='ifInUcastPct',
formula='(100. * ifInUcastPkts / ifInTotalPkts) if ifInUcastPkts else 0.', alert_type=1),
Result(metric_id=8, oid_name='ifInMulticastPct', formula='(100. * ifInMulticastPkts /
ifInTotalPkts) if ifInMulticastPkts else 0.', alert_type=1), Result(metric_id=9,
oid_name='ifInBroadcastPct', formula='(100. * ifInBroadcastPkts / ifInTotalPkts) if
ifInBroadcastPkts else 0.', alert_type=1), Result(metric_id=10, oid_name='ifOutUcastPct',
formula='(100. * ifOutUcastPkts / ifOutTotalPkts) if ifOutUcastPkts else 0.', alert_type=1),
Result(metric_id=11, oid_name='ifOutMulticastPct', formula='(100. * ifOutMulticastPkts /
ifOutTotalPkts) if ifOutMulticastPkts else 0.', alert_type=1), Result(metric_id=12,
oid_name='ifOutBroadcastPct', formula='(100. * ifOutBroadcastPkts / ifOutTotalPkts) if
ifOutBroadcastPkts else 0.', alert_type=1), Result(metric_id=13, oid_name='ifInTotalPkts',
formula='ifInUcastPkts + ifInMulticastPkts + ifInBroadcastPkts + ifInDiscards + ifInErrors',
alert_type=1), Result(metric_id=14, oid_name='ifOutTotalPkts', formula='ifOutUcastPkts +
ifOutMulticastPkts + ifOutBroadcastPkts + ifOutDiscards + ifOutErrors', alert_type=1),
Result(metric_id=15, oid_name='ifInDiscardsPct', formula='(100. * ifInDiscards /
ifInTotalPkts) if ifInDiscards else 0.', alert_type=3), Result(metric_id=16,
oid_name='ifInErrorsPct', formula='(100. * ifInErrors / ifInTotalPkts) if ifInErrors else
0.', alert_type=2), Result(metric_id=17, oid_name='ifOutDiscardsPct', formula='(100. *
ifOutDiscards / ifOutTotalPkts) if ifOutDiscards else 0.', alert_type=3),
Result(metric_id=18, oid_name='ifOutErrorsPct', formula='(100. * ifOutErrors /
ifOutTotalPkts) if ifOutErrors else 0.', alert_type=2), Result(metric_id=19,
oid_name='ifInErrors', formula='ifInErrors', alert_type=2), Result(metric_id=20,
oid_name='ifOutErrors', formula='ifOutErrors', alert_type=2), Result(metric_id=21,
oid_name='ifInDiscards', formula='ifInDiscards', alert_type=3), Result(metric_id=22,
oid_name='ifOutDiscards', formula='ifOutDiscards', alert_type=3), Result(metric_id=23,
oid_name='ifInTotalPct', formula='100. * ifInTotalRate/interfaceSpeed', alert_type=4),
Result(metric_id=24, oid_name='ifOutTotalPct', formula='100. *
ifOutTotalRate/interfaceSpeed', alert_type=4), Result(metric_id=25,
oid_name='ifInUcastRate', formula='ifInUcastPkts/poll_time', alert_type=1),
Result(metric_id=26, oid_name='ifInMulticastRate', formula='ifInMulticastPkts/poll_time',
alert_type=1), Result(metric_id=27, oid_name='ifInBroadcastRate',
formula='ifInBroadcastPkts/poll_time', alert_type=1), Result(metric_id=28,
oid_name='ifOutUcastRate', formula='ifOutUcastPkts/poll_time', alert_type=1),
Result(metric_id=29, oid_name='ifOutMulticastRate', formula='ifOutMulticastPkts/poll_time',
alert_type=1), Result(metric_id=30, oid_name='ifOutBroadcastRate',
formula='ifOutBroadcastPkts/poll_time', alert_type=1), Result(metric_id=31,
oid_name='ifInTotalRate', formula='ifInOctets/(125000*poll_time)', alert_type=4),
Result(metric_id=32, oid_name='ifOutTotalRate', formula='ifOutOctets/(125000*poll_time)',
alert_type=4), Result(metric_id=33, oid_name='ifInOctets', formula='ifInOctets',
alert_type=4), Result(metric_id=34, oid_name='ifOutOctets', formula='ifOutOctets',
alert_type=4)]"
"2020-06-01
12:31:06,197","DEBUG","449","12","collect_if","1685","collect_if.compile_packet_metrics():
l_odefs= [OID(oid_name='ifInBroadcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.3'),
OID(oid_name='ifInDiscards', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.13'),
OID(oid_name='ifInErrors', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.14'),
OID(oid_name='ifInMulticastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.2'),
OID(oid_name='ifInOctets', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.10'),
OID(oid_name='ifInUcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.11'),
OID(oid_name='ifOutBroadcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.5'),
OID(oid_name='ifOutDiscards', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.19'),
OID(oid_name='ifOutErrors', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.20'),
OID(oid_name='ifOutMulticastPkts', snmp_type=0, oid='.1.3.6.1.2.1.31.1.1.1.4'),
OID(oid_name='ifOutOctets', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.16'),
OID(oid_name='ifOutUcastPkts', snmp_type=0, oid='.1.3.6.1.2.1.2.2.1.17'),
OID(oid_name='ifInBroadcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.9'),
OID(oid_name='ifInMulticastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.8'),
```

```
OID(oid_name='ifInOctets', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.6'),
OID(oid_name='ifInUcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.7'),
OID(oid_name='ifOutBroadcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.13'),
OID(oid_name='ifOutMulticastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.12'),
OID(oid_name='ifOutOctets', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.10'),
OID(oid_name='ifOutUcastPkts', snmp_type=1, oid='.1.3.6.1.2.1.31.1.1.1.11')]"

```

stanza:appending task to queue

```
"2020-06-01 12:31:07,336","DEBUG","449","12","shared","139","appending task to queue: 7254"
"2020-06-01 12:31:07,336","DEBUG","449","12","shared","139","appending task to queue: 7260"
"2020-06-01 12:31:07,337","DEBUG","449","12","shared","139","appending task to queue: 10178"

```

The messages will appear in a block before the worker processes begin collecting against individual interfaces.

stanza:got results for

```
"2020-06-01 12:31:08,339","DEBUG","449","12","shared","164","got results for: 7254"
"2020-06-01 12:31:08,340","DEBUG","449","12","shared","164","got results for: 7260"
"2020-06-01 12:31:09,341","DEBUG","449","12","shared","164","got results for: 10178"

```

These messages will appear in a block as the worker processes finish and the main process retrieves results.

stanza:logging stopped

```
"2020-06-01 12:31:10,344","DEBUG","449","12","collect_if","1670","interface collection complete:  processed count: 7, error count 0"
"2020-06-01 12:31:10,346","DEBUG","449","12","proc_pda","528","tracking update: 75 12 c3890be6a40311ea9a0f00221967da01 "
"2020-06-01 12:31:10,349","DEBUG","449","12","proc_pda","545","Stopped tracking process: 75 12 c3890be6a40311ea9a0f00221967da01 "
"2020-06-01 12:31:10,350","DEBUG","449","12","proc_pda","307","Logging stopped for Data Collection: Interface Bandwidth"

```

child process

stanza:starting collect for

```
"2020-06-01 12:30:48,757","DEBUG","31950","12.PoolWorker-10","collect_if","760","ifOperStatus: ('2',) (.1.3.6.1.2.1.2.2.1.8.10104)"
"2020-06-01 12:30:48,781","DEBUG","31950","12.PoolWorker-10","collect_if","760","ifLastChange: ('11681',) (.1.3.6.1.2.1.2.2.1.9.10137)"
"2020-06-01 12:30:48,789","DEBUG","31950","12.PoolWorker-10","collect_if","760","ifInOctets: ('0',) (.1.3.6.1.2.1.31.1.1.1.6.10137)"
"2020-06-01 12:30:48,792","DEBUG","31950","12.PoolWorker-10","collect_if","760","ifAdminStatus: ('2',) (.1.3.6.1.2.1.2.2.1.7.10137)"
"2020-06-01 12:30:48,792","DEBUG","31950","12.PoolWorker-10","collect_if","796","interface_bandwidth.collect(): metric_time: {'ifAdminStatus': 1591014649.0, 'ifOutOctets': 1591014649.0, 'ifLastChange': 1591014649.0, 'ifInOctets': 1591014649.0, 'ifOperStatus': 1591014649.0}"
```

```
"2020-06-01 12:30:48,792","DEBUG","31950","12.PoolWorker-10","collect_if","614","Starting collect for  did:6422 , if_id 476475"
"2020-06-01 12:30:48,793","DEBUG","31950","12.PoolWorker-10","collect_if","704","Collections: set(['ifLastChange', 'ifOutOctets', 'ifAdminStatus', 'ifInOctets', 'ifOperStatus'])"
"2020-06-01 12:30:48,793","DEBUG","31950","12.PoolWorker-10","collect_if","751","PDU packing is disabled"

```

stanza:validating interface

```
"2020-06-01 12:30:50,347","DEBUG","31950","12.PoolWorker-10","collect_if","1086","Validating interface 476475"
"2020-06-01 12:30:50,351","DEBUG","31950","12.PoolWorker-10","collect_if","512","retrieved interface object with id 476475 after querying for last poll of interface 476475"
"2020-06-01 12:30:50,352","DEBUG","31950","12.PoolWorker-10","collect_if","516","This (current poll) interface id = 476475 (<type 'int'>) | Last poll interface id = 476475 (<type 'int'>)"
"2020-06-01 12:30:50,354","DEBUG","31950","12.PoolWorker-10","collect_if","330","IF 476475 - Deriving rates using the following counters:
"2020-06-01 12:30:50,364","DEBUG","31950","12.PoolWorker-10","collect_if","340","IF 476475 - Timestamps (1591013700, 1591014600) - Relationship: TS_MONOTONIC 1 regular timestamps discovered between 1591013700 and 1591014600 - Action: N/A"
"2020-06-01 12:30:50,373","DEBUG","31950","12.PoolWorker-10","collect_if","371","IF 476475 - Last poll and current poll counter relationships and transforms:
"2020-06-01 12:30:50,378","DEBUG","31950","12.PoolWorker-10","collect_if","404","IF 476475 - Regularized counter values and derived rates:
"2020-06-01 12:30:50,383","DEBUG","31950","12.PoolWorker-10","collect_if","1171","Using 10000000 as interface speed."
"2020-06-01 12:30:50,384","DEBUG","31950","12.PoolWorker-10","collect_if","1419","State change alerting for ifOperStatus"
"2020-06-01 12:30:50,384","DEBUG","31950","12.PoolWorker-10","collect_if","1419","State change alerting for ifAdminStatus"
"2020-06-01 12:30:50,384","DEBUG","31950","12.PoolWorker-10","collect_if","1250","Total_in: None "
"2020-06-01 12:30:50,384","DEBUG","31950","12.PoolWorker-10","collect_if","1261","Total_out: None "
"2020-06-01 12:30:50,385","DEBUG","31950","12.PoolWorker-10","collect_if","1288","New threshold info: InUni: None InNUni: None  ,Total: None Perc: 0.0 & 0.0"
"2020-06-01 12:30:50,385","DEBUG","31950","12.PoolWorker-10","collect_if","1289","New threshold info: OutUni: None OutNUni: None, Total: None Perc: 0.0 & 0.0"
"2020-06-01 12:30:50,386","DEBUG","31950","12.PoolWorker-10","collect_if","1387","Threshold evaluation on DID 6422/IF_ID 476475. Clearing Alerts: []. Exceeded thresholds: []. Active Alerts: set([]). Current Poll: {'ifInTotalRate': 0.0, 'ifInTotalPct': 0.0, 'ifOutTotalPct': 0.0, 'ifOutTotalRate': 0.0}. Previous poll: {'ifInTotalRate': 0, 'ifInTotalPct': 0, 'ifOutTotalPct': 0, 'ifOutTotalRate': 0}."
"2020-06-01 12:30:50,389","DEBUG","31950","12.PoolWorker-10","interface","761","Getting flexible if_name from if_id and did: 476475 - 6422"
"2020-06-01 12:30:50,391","DEBUG","31950","12.PoolWorker-10","database","173","QL: query: SELECT SQL_CACHE xid,xtype,format FROM master_dev.device_interfaces_name_overrides
"2020-06-01 12:30:50,392","DEBUG","31950","12.PoolWorker-10","database","174","QL: dbc id: 140312011380816"
"2020-06-01 12:30:50,394","DEBUG","31950","12.PoolWorker-10","database","175","QL: query time: 0.00128412246704"
"2020-06-01 12:30:50,395","DEBUG","31950","12.PoolWorker-10","database","176","QL: lastrowid:  0"
"2020-06-01 12:30:50,396","DEBUG","31950","12.PoolWorker-10","database","177","QL: rowcount: 0"
```

```
"2020-06-01 12:30:50,396","DEBUG","31950","12.PoolWorker-10","interface","770","No matching interfaces override format found for interface '476475'.Get from system settings,"
"2020-06-01 12:30:50,410","DEBUG","31950","12.PoolWorker-10","interface","785","Interface Name Format is: {name}"
"2020-06-01 12:30:50,412","DEBUG","31950","12.PoolWorker-10","database","173","QL: query: SELECT SQL_CACHE name,name,alias
"2020-06-01 12:30:50,413","DEBUG","31950","12.PoolWorker-10","database","174","QL: dbc id: 140312011380816"
"2020-06-01 12:30:50,413","DEBUG","31950","12.PoolWorker-10","database","175","QL: query time: 0.00177383422852"
"2020-06-01 12:30:50,413","DEBUG","31950","12.PoolWorker-10","database","176","QL: lastrowid:  0"
"2020-06-01 12:30:50,414","DEBUG","31950","12.PoolWorker-10","database","177","QL: rowcount: 1"
"2020-06-01 12:30:50,414","DEBUG","31950","12.PoolWorker-10","interface","800","Before parsing Interface Name Format {name}"
"2020-06-01 12:30:50,417","DEBUG","31950","12.PoolWorker-10","collect_if","550","inserting pickled interface 476475"

```

stanza:getting flexible if_name

```
"2020-06-01 12:30:51,927","DEBUG","31950","12.PoolWorker-10","interface","761","Getting flexible if_name from if_id and did: 476475 - 6422"
"2020-06-01 12:30:51,927","DEBUG","31950","12.PoolWorker-10","database","173","QL: query: SELECT SQL_CACHE xid,xtype,format FROM master_dev.device_interfaces_name_overrides
"2020-06-01 12:30:51,928","DEBUG","31950","12.PoolWorker-10","database","174","QL: dbc id: 140312011380816"
"2020-06-01 12:30:51,928","DEBUG","31950","12.PoolWorker-10","database","175","QL: query time: 0.000690937042236"
"2020-06-01 12:30:51,928","DEBUG","31950","12.PoolWorker-10","database","176","QL: lastrowid:  0"
"2020-06-01 12:30:51,929","DEBUG","31950","12.PoolWorker-10","database","177","QL: rowcount: 0"
"2020-06-01 12:30:51,929","DEBUG","31950","12.PoolWorker-10","interface","770","No matching interfaces override format found for interface '476475'.Get from system settings,"
"2020-06-01 12:30:51,930","DEBUG","31950","12.PoolWorker-10","interface","785","Interface Name Format is: {name}"
"2020-06-01 12:30:51,931","DEBUG","31950","12.PoolWorker-10","database","173","QL: query: SELECT SQL_CACHE name,name,alias
"2020-06-01 12:30:51,932","DEBUG","31950","12.PoolWorker-10","database","174","QL: dbc id: 140312011380816"
"2020-06-01 12:30:51,932","DEBUG","31950","12.PoolWorker-10","database","175","QL: query time: 0.000500202178955"
"2020-06-01 12:30:51,932","DEBUG","31950","12.PoolWorker-10","database","176","QL: lastrowid:  0"
"2020-06-01 12:30:51,932","DEBUG","31950","12.PoolWorker-10","database","177","QL: rowcount: 1"
"2020-06-01 12:30:51,933","DEBUG","31950","12.PoolWorker-10","interface","800","Before parsing Interface Name Format {name}"
"2020-06-01 12:30:51,933","DEBUG","31950","12.PoolWorker-10","collect_if","1456","Interface Name from format Gi0/5"

```

single:about to save raw stats

```
"2020-06-01 12:30:52,193","DEBUG","31950","12.PoolWorker-10","collect_if","1474","About to save raw stats did: 6422, if_id: 476475, collect time: 1591014600"
```

```
2
```

Questions

See here for a guide to adding to this section: [section: questions]

Orphaned Interfaces (did=0)

Details

Why do these occur?

- If Nightly Discovery detects that an interface is no longer associated with a device, that interface gets set to did=0 in order to preserve the historical data in case the interface comes back again

What is the right way to delete did=0 interfaces from a customer system so that linked data is also deleted?

- First identify the need for the request to do so. Is there a problematic behavior believed to be tied to these entries? There exists backend process logic that depend on did=0 for interfaces. For example, Nightly Discovery will try to check those "`orphaned"` interfaces in order to update device interfaces as will the `topology  crunch` process, etc...
  Ex. SL1 does a query to find previously discovered interfaces that have moved devices. The large number of orphaned entries cause the query to match in places where it should not and pull these entries out of orphan status.
- You will also want to identify over what time span these accrued. Is this over a single day, or over a long period of time?
  but it is a concern that so many interfaces got the did = 0 .
  are these interfaces all from one day's change, or accumulated over a period?
  also, are they running into certain issue that they have to clean up those interfaces?

- **Cleanup**: **[Warning]** Performing mass deletes where did=0 in `master_dev`.`device_interfaces` may result in loss of history for interfaces that have been move from one device to another and interfaces that may be temporarily down. Any manual deletions should be done precisely rather than deleting WHERE did=0

Commentary

- 2019-10-07 Bryan Harding
  - I'm getting pinged by Support asking about ways for cleaning up what they are reporting as "orphaned" interfaces in master_dev.device_interfaces with did = 0. Looking through JIRA and Slack, I can see references suggesting that did=0 when the interface is removed. Nearest defect I can find is EM-3301 fixed way back in 7.8.1, while this customer is on 8.10.0
  - Why wouldn't these get cleaned up automatically?